

Contextual Metadata

Instagram Profile Analysis Dashboard

Dataset Overview

This project is based on one year of Instagram data from my sister's account. The analysis focuses on three areas: the posts and reels she liked, the accounts following her, and the accounts she follows.

Only the files needed for these analyses were selected from the Instagram export. Information such as private messages, account settings, advertising data, security information, and media files was not used.

The original files were kept unchanged in the 'raw_data' folder. The resulting dashboard shows overall activity, liking patterns, and changes in followers and following over time.

Raw Data Files

1. liked_posts.json

This file contains records of Instagram posts and reels that I liked during the selected period. Each record includes a timestamp, content ID, and information related to the Instagram content URL. The URL was used to identify whether the liked content was a Post or a Reel.

The Unix timestamps were converted into normal date and date-time values in Power Query. Additional fields were created to analyze the activity by date, hour, and time period. The cleaned data was used to create the FactLikes table.

2. followers_1.json

This file contains information about Instagram accounts that followed me. Each record contains a follower username and a Unix timestamp showing when the follow activity occurred.

The timestamps were converted into date and date-time values in Power Query. The cleaned records were used to create the FactFollowers table, which is used for follower totals and monthly follower analysis.

3. following.json

This file contains information about Instagram accounts that I follow. The raw data includes usernames, Instagram profile links, and timestamps related to the following activity.

The timestamps were converted into date and date-time values in Power Query. The cleaned data was used to create the FactFollowing table and is used to analyze the number of accounts followed and changes over time.

Power BI Tables

- FactLikes: Contains my Instagram liking activity and is used to calculate Total Likes, Total Posts, Total Reels, Reel Like %, and monthly changes.
- FactFollowers: Contains information about my followers and is used to calculate Total Followers.
- FactFollowing: Contains information about the accounts I follow and is used to calculate Total Following.
- DimDate: Contains dates, months, years, weekdays, and quarters. It is used for analyzing activity over time.
- DimContent: Groups liked content into Posts and Reels.
- DimTime: Groups activity into Morning, Afternoon, Evening, and Night.
- ContentSummary: A calculated table that summarizes likes by content type.

Analytical Model

The data was organized into separate tables based on the type of Instagram information being analyzed. FactLikes stores the liked-content activity, while FactFollowers and FactFollowing contain the audience-related records.

To make filtering and analysis easier, three supporting tables were added. DimDate provides the calendar information used across the activity tables, DimContent separates liked items into posts and reels, and DimTime groups activity according to the part of the day. DimDate was created in Power Query using M.

The tables are connected so that the same date filters can be used when analyzing likes, followers, and following. A separate ContentSummary table was also created with DAX to provide a simple summary of likes for each content type.

Dashboard Structure

Instagram Overview presents the main KPIs and overall liking activity. It includes Total Likes, Total Posts, Total Reels, Reel Like %, content-type analysis, monthly activity, and a monthly summary.

Activity Analysis focuses on when Instagram liking activity occurs. It analyzes likes by hour, weekday, and time period and allows the data to be filtered by date, content type, and time period.

Audience Analysis focuses on followers and following activity. It includes Total Followers, Total Following, the Follower/Following Ratio, and monthly trends for new followers and accounts followed.